

Nikhil C R

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Professional Summary

AI Engineer specializing in Computer Vision and Generative AI, with hands-on experience building robust end-to-end solutions—from research agents and interview simulators to advanced RAG pipelines. Adept at crafting high-performing vision models using state-of-the-art models. Proficient in PyTorch, OpenCV, LangGraph, and vector databases, with a proven ability to transform research into real-world, application-ready AI tools.

Experience

Junior AI Engineer, Ombrulla May 2025 – Present

- Developed and deployed AI-powered visual inspection pipelines to automate defect detection and quality control reducing manual review time and improving throughput.
- Designed and implemented Retrieval-Augmented Generation (RAG) systems to enhance document search workflows, improving information relevance and response quality.
- Fine-tuned and optimized object detection and segmentation models for industrial scenarios with challenging lighting, perspective, and defect variability.
- Built modular, reusable pipelines for data ingestion, preprocessing, model training, and inference.

Intern, Optimization and Machine Learning Lab, Digital University Kerala Feb 2024 – July 2024

- Developed a few-shot Object Detection model using Faster R-CNN to accurately detect road damage.
- Leveraged Detectron2 and PyTorch to build the model.
- Implemented a Two-stage Fine-tuning Approach for improving detection performance on under-represented damage categories.

Education

- **Kerala University of Digital Sciences, Innovation and Technology** 2022 – 2024
Master of Science in Computer Science with specialization in Machine Intelligence
- **University of Calicut** 2019 – 2022
Bachelor of Science in Computer Science

Projects

- **AI Web Research Agent**
 - Developed a research system utilizing LangGraph to autonomously analyze complex queries, formulate dynamic research strategies, and synthesize information from multiple sources.
 - Integrated and orchestrated various external tools, utilized gemini api for query analysis, Tavily for web search, News API for news retrieval, and utilized crawl4ai for web scraping.
- **Agent Driven Interview Simulation**
 - Developed a comprehensive system to simulate technical interviews, integrating gemini for dynamic question selection, candidate response analysis, and automated feedback generation.
 - Implemented robust conversational flow and state transitions using the LangGraph.
 - Designed and implemented the API architecture to ensure seamless integration with the frontend

- **Llama based Mental Health Counseling Assistant**

- Developed a specialized Mental Health Counseling Assistant by fine-tuning Meta’s Llama 3.1 (8B) to provide structured advice in a consistent format.
- Processed a dataset of counseling conversations for generating synthetic samples required for the fine-tuning.
- Implemented efficient fine-tuning using the QLoRA approach with quantization

- **Aviation Regulations RAG System**

- Developed and implemented a Retrieval-Augmented Generation (RAG) system to enable querying of complex Aviation Regulations.
- Designed and built the system for eCFR API fetching, XML parsing, document chunking, and generating vector embeddings using OpenAI models.
- Integrated ChromaDB for efficient storage and retrieval, enhanced by Cohere reranking for improved search relevance.
- Created an interactive web application using Streamlit

- **Road Damage Detection Using Faster R-CNN**

- Developed an Object Detection model using Faster R-CNN to accurately detect road damage.
- Leveraged Detectron2 and PyTorch to build the model.
- Implemented Two-stage Fine-tuning Approach for improving detection performance on under-represented damage categories.

Awards and Recognitions

- Winner of the **AI Filmmaking Competition** at the second edition of **100xEngineers Generative AI Buildathon** June 2025
- Winner of the **AI Short Film Competition** at the **100xEngineers Generative AI Buildathon** December 2024
- Winner of the **INNOFEST Hackathon** held at the **Kerala University of Digital Sciences, Innovation and Technology** September 2023
- Received the award for **Best Contribution to Corporate Life of Prajyoti Niketan College** March 2022

Certifications

- **Data Structures and Algorithms in Python**
jovian.ai
- **JavaScript Algorithms and Data Structures**
freecodecamp.org

Skills

- **Technical Skills:**

- **Languages:** Python, Javascript, SQL
- **AI/ML:** Deep Learning, Computer Vision, Generative AI, Object Detection, LLM Fine-tuning, Retrieval Augmented Generation (RAG), Agentic Systems, Transfer Learning
- **Frameworks & Libraries:** PyTorch, Detectron2, LangGraph, Huggingface-Transformers, Langchain, YOLO, OpenCV, Pandas, NumPy

- **Soft Skills:** Problem-solving, Collaboration & Teamwork, Communication, Analytical Thinking